

Touch-Screen, Wireless and Detachable Displays Make Modern Multimeters Exciting



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From general-purpose to high-end measurements, multimeters are widely-used tools. Today, there is an extensive variety of multimeters in the market, including ones with wireless capabilities and touch-screen interfaces to make measurements intuitive. In this article, we attempt to guide you in selecting the right multimeter for your needs.

Also known as multitesters or volt-ohm meters, multimeters are offered by various test and measurement (T&M) companies in diverse form factors, from benchtop to handheld and clamped. These fit a large array of applications, such as industrial, educational or general-purpose measurement, to name a few.

Whether you are looking to buy a new multimeter or a used one, it is important to understand the capabilities as well as performance of the device. Accuracy is an essential factor that could determine the choice of the multimeter you plan to settle for. Depending on your usage and application, there is a plethora of multimeters available in the market. Let us dive into how T&M vendors are improving these devices feature-wise, which will be followed by some impor-

tant parameters that need to be considered before buying a multimeter. Along the way, we also highlight some of the latest entrants in multimeters.

Trends in today's multimeters—data logging and wireless connectivity

Increasingly, digital multimeters need to offer more multifunction capabilities to suit a wider scope of applications in today's demanding measurement environments. Yet, the multimeter preferably needs to be more compact and lighter than before, and be able to deliver higher accuracy across a larger number of measurements, while continuing to offer safety, warranty and support, informs Ravi Pagar, regional director, element14. He says, "In recent years, data sharing is a key requirement, where users need to share a large amount of data quickly and accurately." Latest improvements in digital multimeters, which include addition of more useful functions, connectivity and higher safety levels, are making these more versatile.

Multimeters these days are sturdier and electrically safe for the person handling these, informs Kritikesh Kumar, national sales and marketing head - test and measurement business division, Yokogawa India Ltd. He says, "Apart from the ruggedness, digital multimeters churn out a much better level of accuracy and resolution. Users can now be more sure about the readings obtained in their instruments." He adds, "Multimeters also come with data logging and PC connectivity. This cuts down the task of manually writing down data or finding a place to store it."

Nowadays, wireless connectivity is also increasingly being incorporated into multimeters. Industry experts feel that connectivity today is slowly becoming a part of multimeters. Manish Kwatra, CEO,

Fig. 1: Keysight's U1200 series digital multimeters, where U1117A IR-to-Bluetooth adaptor can be connected to the IR port for enabling wireless connectivity

